First, I opened my Ubuntu terminal in VirtualBox and navigated to my xv6 project folder: I ran: cd ~/os_hw2/xv6-riscv

Before starting the implementation, I reviewed the xv6 sleep lock code to understand how locks are used in the operating system. I also reviewed the differences between spin locks and sleep locks and studied the Anderson array based queue lock. I also studied the Anderson array based queue lock .

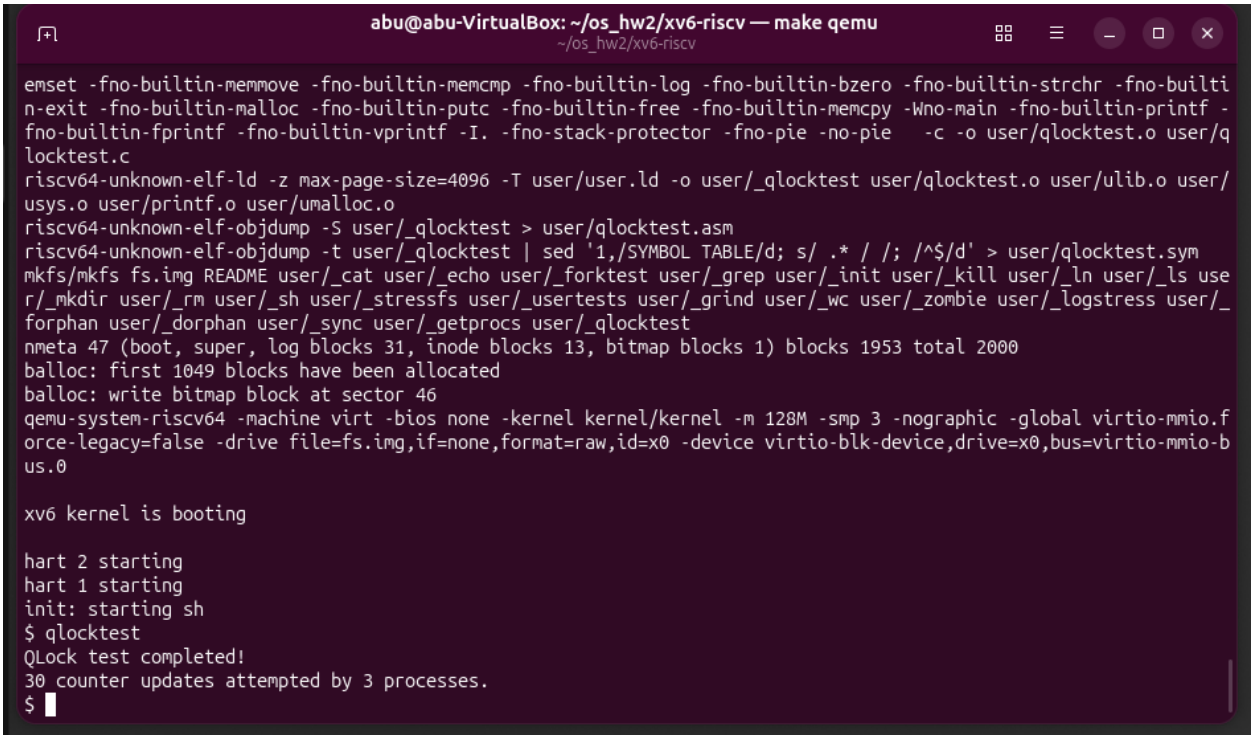
Afterward I created two files for the QLock implementation, kernel/qlock.h and kernel/qlock.c. The header file contains the QLock structure, the lock states and the function declarations. The QLock used two main states the HAS_LOCK which means the slot currently has the lock. The MUST_WAIT which means the slot must wait until the lock becomes available. The implementation also uses an array of flags and a tail variable to keep track of the next position in the queue. I created three main functions, initqlock(): which Initializes the QLock and sets the starting lock state. acquireq(): which requests the lock and waits until the assigned slot receives permission to continue. releaseq(): which releases the current slot and gives the lock to the next slot in the queue.

After creating the QLock files, I connected the implementation to the xv6 kernel. I modified the these files:

- kernel/main.c, which Initializes the QLock when xv6 starts.
- kernel/sysproc.c, which contains the system call that uses the QLock.
- kernel/syscall.c, which connects the system call to the kernel.
- kernel/syscall.h, which defines the system call number.
- Makefile, which includes the new QLock source file and test program.
- user/user.h, which declares the user level system call.
- user/usys.pl, which creates the system call entry.
- user/qlocktest.c, which contains the test program.

I also created a system call called qlocktest to test the QLock using a shared counter and after connecting everything, I rebuilt xv6 using this command: make clean and make qemu. The system compiled successfully and started without errors and once xv6 started, I ran the test program qlocktest, and the program created three child processes. Each child process calls the QLock test system call ten times. Inside the system call, the QLock protects a shared counter. Each time a process gets the lock, it increases the counter by one and then releases the lock.

The test completed successfully and the output showed the counter increasing from 1 to 30. This matches the three child processes making ten counter updates each. The program also displayed the message “QLock test completed”, and the test showed that the QLock protected the shared counter during this test with multiple processes.


abu@abu-VirtualBox: ~/os_hw2/xv6-riscv — make qemu
~/os_hw2/xv6-riscv
emset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builti
n-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -
fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/qlocktest.o user/q
locktest.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_qlocktest user/qlocktest.o user/ulib.o user/
usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_qlocktest > user/qlocktest.asm
riscv64-unknown-elf-objdump -t user/_qlocktest | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/qlocktest.sym
mkfs/mkfs fs.img README user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls use
r/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_
forphan user/_dorphan user/_sync user/_getprocs user/_qlocktest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1049 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.f
orce-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-b
us.0
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ qlocktest
QLock test completed!
30 counter updates attempted by 3 processes.
$

```
user/_dorphan user/_sync user/_getprocs user/_qlocktest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1049 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force
-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ qlocktest
Counter value: 1
Counter value: 2
Counter value: 3
Counter value: 4
CCCoouunter value: 5
Counter value:
 8
Counter value: 9
Counnter value: 6
Counter value: 11
Counter ter value:value: 12
Counter value 10
Counte:r value :13
Counter 14
Counter value:  16
valueoun: 15
Counter value:ter v 1a7l
ueC: ou7
Counter value: 19
Counter value: 20
Counter value: 21
Counter valnute:e r2 2
vaClue: 18
Counter value: 24
Countero value: 25
Countuenrt value: 26
er value: 23
Counter value: 27
Counter value: 28
Counter value: 29
Counter value: 30
QLock test completed!
30 counter updates attempted by 3 processes.
$ $
```